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Pullbacks, universal properties, and mathematical writing

A development specimen for mathematical reading: an elementary pullback, its universal property, carefully linked statements, and native TikZ-CD diagrams.

This is a **development specimen**, not a published article by Vihaan Dheer. The elementary examples below exercise the reading layout and authoring tools. They are deliberately more varied than a short blog post would usually need. For a quieter version of the same idea, see A small observation about pullbacks.

A useful way to understand a construction is to ask which maps it represents. For abelian groups A, B, and C with homomorphisms $f\colon A \to C$ and $g\colon B \to C$, the pullback $A \times_C B$ remembers a pair whose two images agree. The content is simple; the notation has to remain comfortable even in a paragraph containing $\mathrm{Hom}_{\mathbb{Z}}(T, A \times_C B)$, a script letter $\mathscr{L}$, a calligraphic sheaf $\mathcal{E}$, or a quotient $\frac{|x|+\frac{1}{2}}{1+|x|}$. These belong to the same line of thought.

The forward link to Theorem 2 already has its final target. The displayed identity (2) is collected before any reference is rendered; it is not a number typed into this paragraph.

A square and its objects

Definition 1 (Pullback of abelian groups).

For $f\colon A \to C$ and $g\colon B \to C$, define

$$A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\}. \tag{1}$$

The projection maps are $p_A(a, b) = a$ and $p_B(a, b) = b$.

This is a subgroup of $A \times B$: equality of the two images is preserved by addition and additive inverses. In particular, the notation does not add a new choice to the construction. The projections fit into Diagram 1.

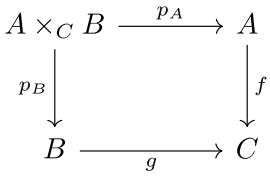


Diagram 1. The defining square for $A \times_C B$.

The pullback P = A times over C B maps to A by p_A and to B by p_B. The maps f from A to C and g from B to C satisfy f p_A = g p_B. The square is commutative.

▶ Diagram TeX source

Both the paragraph and the diagram use the same three-argument `\pullback` command. The diagram is compiled from its retained TeX source by TikZ-CD; it is not a substitute drawn by the page layout. Its description records the objects, arrows, and equality even when its path-based glyphs are not read as text.

For a small square, the ordinary equation renderer also supports a simple CD:

$$\begin{array}{ccc} A\times_C B & \xrightarrow{p_A} & A \\ {\scriptstyle p_B}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow[g]{} & C \end{array}$$

This lighter syntax is useful on occasion, but the native route above is the one that supplies the full arrow vocabulary.

The universal property of Hom

Theorem 2 (Maps into a pullback).

For every abelian group T, the projections induce a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathbb{Z}}(T, A \times_C B) &\longrightarrow \mathrm{Hom}_{\mathbb{Z}}(T, A) \times_{\mathrm{Hom}_{\mathbb{Z}}(T,C)} \mathrm{Hom}_{\mathbb{Z}}(T, B), \\ u &\longmapsto (p_A u, p_B u). \end{aligned} \tag{2}$$

This bijection is natural in T.

Proof (of the universal property).

A homomorphism u gives a compatible pair because the square commutes. Conversely, suppose $a\colon T \to A$ and $b\colon T \to B$ satisfy $fa = gb$. Then $t \mapsto (a(t), b(t))$ takes values in the subgroup in Definition 1 and is a homomorphism. Its two projections are a and b, so it is the only possible map with those projections.

Precomposition by $v\colon T' \to T$ takes the pair (a, b) to (av, bv) on both sides of (2), proving naturality.

□

We have proved the universal property directly, without choosing generators. The argument works for maps of sets as well; the extra group structure only asks us to check that the mediating function is a homomorphism.[1]

Lemma 3 (An injectivity test).

If f is injective, then $p_B\colon A \times_C B \to B$ is injective.

Proof.

If $p_B(a, b) = 0$, then $b = 0$ and $f(a) = g(0) = 0$. Injectivity of f gives $a = 0$, as required.

□

Proposition 4 (Pulling back an isomorphism).

If f is an isomorphism, then p_B is an isomorphism, with inverse $b \mapsto (f^{-1}g(b), b)$.

Proof.

The displayed map is a homomorphism into the pullback. Its composite with p_B is the identity. In the other direction, every (a, b) in the pullback satisfies $f^{-1}g(b) = a$. These equalities prove both inverse identities.

□

Corollary 5 (The identity case).

Taking $A = C$ and $f = \mathrm{id}_C$ gives $C \times_C B \cong B$ canonically.

The word *canonically* has a concrete meaning here: the isomorphism is the projection already present in Diagram 1. It does not mean an unspecified isomorphism obtained by comparing cardinalities.

Examples, notation, and long expressions

Example 6 (Two integer multiplications).

Let $A = B = C = \mathbb{Z}$, with $f(a) = 2a$ and $g(b) = 3b$. A compatible pair has the form $(a, b) = (3k, 2k)$ for a unique integer k. Thus this pullback is isomorphic to $\mathbb{Z}$ by $k \mapsto (3k, 2k)$.

Remark 7 (Equality and isomorphism).

The set in (1) is a particular model. An isomorphic model has the same universal property, but it need not have the same elements. We keep $=$, $\cong$, and $\simeq$ distinct instead of silently exchanging one arrow or relation for another.

Question 8 (A small exercise).

If $f\colon A \to C$ is surjective, is p_B surjective for every $g\colon B \to C$? Use the description of a compatible pair to give a direct proof.

Remark (Notation in this specimen).

The shared commands $\mathbb{Z}$, Hom, $|x|$, and $\langle x, y \rangle$ work in article math and native diagrams. The post deliberately overrides $\mathcal{E}$ with a new calligraphic letter using `\renewcommand`. That override does not affect A small observation about pullbacks.

The following alignment tests nested fractions, long subscripts, scripts, blackboard letters, and accents. It is a notation exercise, not an asserted identity involving the preceding pullback:

$$
\begin{aligned}
q(x) &:= \frac{1 + \dfrac{|x|^2}{1 + |x|}}{2 + |x|}, \qquad x \in \mathbb{R}, \\
V_{\substack{0 \le i \le n \\ j \in I_{\text{finite}}}} &:= \bigoplus_{\substack{0 \le i \le n \\ j \in I_{\text{finite}}}} \mathbb{Q} e_{i,j}, \\
\hat{\mathcal{E}}, &\quad \tilde{\mathcal{G}}, \quad \mathscr{L}, \quad \mathfrak{m} \qquad \text{(four distinct notation samples)}.
\end{aligned}
$$

Authors can normally split a long equality at a meaningful relation. This single unbroken product is retained to test local keyboard and touch scrolling:

$$
\underbrace{\operatorname{Hom}_{\mathbb{Z}}(T, A) \times_{\operatorname{Hom}_{\mathbb{Z}}(T, C)} \operatorname{Hom}_{\mathbb{Z}}(T, B)}_{\text{compatible pairs of maps from } T} \times \underbrace{\operatorname{Hom}_{\mathbb{Z}}(T', A) \times_{\operatorname{Hom}_{\mathbb{Z}}(T', C)} \operatorname{Hom}_{\mathbb{Z}}(T', B)}_{\text{compatible pairs of maps from } T'}. \tag{3}
$$

The equation number remains available while its expression scrolls. This should never make the surrounding page wider than the phone.

A native arrow vocabulary

The second native specimen is a **syntax gallery**, not a commutative diagram asserting relationships between particular objects. Its curved and diagonal arrows, parallel pair, hook, surjection, dashed arrow, and equality have different meanings; the source preserves those differences.

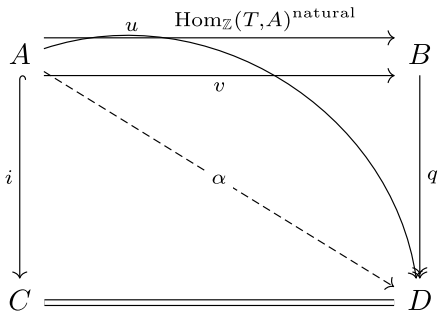


Diagram 2. Native TikZ-CD arrow and label specimen; no commutativity is asserted.

A syntax gallery on objects A, B, C, and D: A has two parallel arrows to B, a hook to C, and a diagonal dashed arrow to D. B has a two-headed arrow to D. C has an equality arrow to D. A curved arrow from A to D carries a long superscripted label. This gallery tests rendering and asserts no commutativity.

▶ Diagram TeX source

For further details about the construction itself, the Stacks Project's definition of a fibre product is a useful reference. The discussion here is original elementary exposition; the linked sources supply terminology and broader context.

References

1. The Stacks Project Authors, *The Stacks Project*, Fibre products and pushouts, Tag 001U.
2. The Stacks Project Authors, *The Stacks Project*, Total direct image, Tag 08DY, consulted as a reference for mathematical reading and navigation.
3. Jacob Lurie, *Kerodon*, The Language of ∞-Categories, Tag 0001, consulted as a typography and cross-reference reference.

Footnotes

1. The naturality assertion concerns precomposition in T. It says the bijections commute with the maps induced by $v \colon T' \to T$, rather than merely that each bijection exists. ↩

This article is a development specimen for the reading and authoring system. It is excluded from the published site.